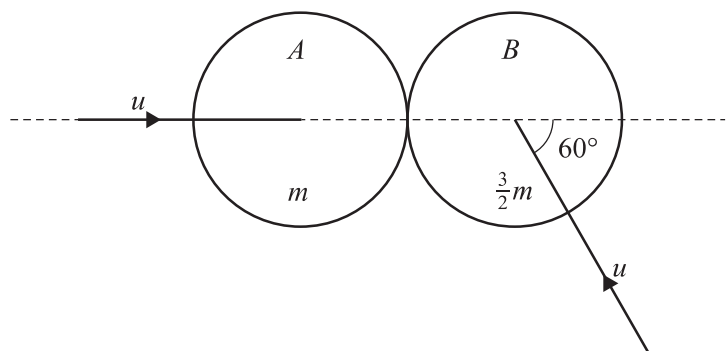


Find the value of  $k$ . [6]

This image shows a full page of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page, providing a template for handwriting practice or general note-taking. There are no margins, text, or other markings on the page.



(a) Find the angle through which the direction of motion of  $B$  is deflected by the collision. [6]

[illegible]

- [illegible]